Agent Commerce Gateway (ACG / MACCP)

Evaluator One-Page Summary | Razorpay AI Buildathon — Track 01

> ***“AI proposes. ACG authorizes. Razorpay executes.”***

> ***“The model can propose anything. It cannot authorize anything.”***

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1. Problem & Positioning

Autonomous AI agents can formulate carts and trigger payments, but merchants cannot allow untrusted LLMs to dictate prices, overstep budgets, or race against inventory. ACG sits as an optional **merchant-side control plane** between upstream agents (ChatGPT, Claude, Gemini, Cursor) and downstream payment execution (Razorpay).

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2. Core Security & Control Pipeline

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ANY AI AGENT ==> ADAPTER REGISTRY ==> CANONICAL INTENT ==> MANDATE CHECK (Ed25519)
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MERCHANT TRUTH (DB Price) ==> POLICY ENGINE ==> DUAL ATOMIC LOCK ==> RAZORPAY EXECUTION
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3. Verified Empirical Results

Metric	Target	Verified Value	Status
Automated Test Suite	100% Pass Rate	**77 / 77 Tests Passed** (9 Test Suites)	**PASS**
Adversarial / Security Pentest	Zero Breaches	**19 / 19 Live HTTP Vectors Blocked**	**PASS**
Unauthorized Financial Paths	Zero Tolerance	**0 Paths Observed**	**PASS**
Cold-Start Transaction Speed	Sub-500 ms SLA	**303.81 ms** End-to-End Order Creation	**PASS**
Merchant Onboarding Velocity	Rapid Deployment	**10–12 minutes** Configuration Setup	**PASS**
Audit Ledger Provenance	Tamper-Evident	**307 SHA-256 Chained Blocks Verified**	**PASS**

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4. Ecosystem Interoperability

* `LIVE`: Native ACG, REST Ingress, Razorpay Sandbox API.

* `ADAPTER READY`: MCP (`tools/call`), A2A (Linux Foundation), ACP (1.0), AP2 (0.2.0), UCP (1.2).

* `DESIGN`: Visa TAP (Hardware Enclaves).

* `ADVISORY`: Razorpay Vulcan Intelligence (Downstream risk and optimal rail modeling).

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5. Architectural Boundary & Release Status

* **Status: PASS WITH OBSERVATIONS** (Verified Sandbox Candidate).

* **Boundary:** Single-node SQLite reference implementation verified for ACID transaction atomicity. Multi-instance production roadmap documented for PostgreSQL and Redis distributed locking.